# Introduction to Electronic Design Automation
# National Taiwan University
# Spring 2025

## Problem Set 2
## Reference Solution

## 1 [Cofactor] (9%)

W.L.O.G. assume that $v$ is a positive literal.

### (a) (3%)

The statement is false. Let $f(x) = x$, $u = \neg x$, and $v = x$. We have

$$(f_u)_v = (0)_v = 0\,,$$

and

$$(f_v)_u = (1)_u = 1\,.$$

### (b) (3%)

The statement is true. Let $f = f(v, W)$. We have

$$(\neg f)_v = (\neg f(v, W))_v = \neg f(1, W) = \neg f_v\,.$$

### (c) (3%)

The statement is true. Let $f = f(v, W)$ and $g = g(v, W)$. We have

$$(f \oplus g)_v = (f(v, W) \oplus g(v, W))_v = f(1, W) \oplus g(1, W) = f_v \oplus g_v\,.$$

## 2 [Quantified Boolean Formula] (15%)

### (a) (3%)

The biconditional holds. Note that

$$\begin{aligned}
\neg(\forall x, \exists y. f(x, y, z)) &= \neg(\forall x.(f(x, 0, z) \vee f(x, 1, z))) \\
&= \neg((f(0, 0, z) \vee f(0, 1, z)) \wedge (f(1, 0, z) \vee f(1, 1, z))) \\
&= (\neg(f(0, 0, z) \vee f(0, 1, z)) \vee \neg(f(1, 0, z) \vee f(1, 1, z)) \\
&= ((\neg f(0, 0, z) \wedge \neg f(0, 1, z)) \vee (\neg f(1, 0, z) \wedge \neg f(1, 1, z))\,,
\end{aligned}$$

and

$$\begin{aligned}
\exists x, \forall y. \neg f(x, y, z) &= \exists x.(\neg f(x, 0, z) \wedge \neg f(x, 1, z)) \\
&= (\neg f(0, 0, z) \wedge \neg f(0, 1, z)) \vee (\neg f(1, 0, z) \wedge \neg f(1, 1, z)) \\
&.
\end{aligned}$$

**(b) (3%)**

A counterexample for the ($\Rightarrow$) direction is $f = x$. We have

$$\forall x, \exists y. f(x, y, z) = \forall x. x = 0\,,$$

while

$$\exists x, \forall y. f(x, y, z) = \exists x. x = 1\,.$$

A counterexample for the ($\Leftarrow$) direction is $f = y$. We have

$$\forall x, \exists y. f(x, y, z) = \forall x.(1) = 1\,,$$

while

$$\exists x, \forall y. f(x, y, z) = \exists x.(0) = 0\,.$$

**(c) (3%)**

The ($\Rightarrow$) direction is correct. To see this, note that

$$\begin{aligned}\exists x, \forall y. f(x, y, z) &= \exists x.(f(x, 0, z) \wedge f(x, 1, z)) \\ &= (f(0, 0, z) \wedge f(0, 1, z)) \vee (f(1, 0, z) \wedge f(1, 1, z)) \\ &= (f(0, 0, z) \vee f(1, 0, z)) \wedge (f(0, 1, z) \vee f(1, 1, z)) \\ &\quad \wedge (f(0, 0, z) \vee f(1, 1, z)) \wedge (f(0, 1, z) \vee f(1, 0, z))\,,\end{aligned}$$

and

$$\begin{aligned}\forall y, \exists x. f(x, y, z) &= \forall x.(f(0, y, z) \vee f(1, y, z)) \\ &= (f(0, 0, z) \vee f(1, 0, z)) \wedge (f(0, 1, z) \vee f(1, 1, z))\,,\end{aligned}$$

which is clearly implied by the first one. A counterexample for the ($\Leftarrow$) direction is $f = x \oplus y$. We have

$$\exists x, \forall y. f(x, y, z) = \exists x.(x \oplus 0) \wedge (x \oplus 1) = \exists x.(\neg x \wedge x) = \exists x.(0) = 0\,,$$

while

$$\forall y, \exists x. f(x, y, z) = \forall y.(0 \oplus y) \vee (1 \oplus y) = \forall y.(y \vee \neg y) = \forall y.(1) = 1.$$

**(d) (3%)**

The biconditional holds. Note that

$$\forall x.(f(x, y) \wedge g(x, z)) = f(0, y) \wedge g(0, z) \wedge f(1, y) \wedge g(1, z)\,,$$

and

$$\begin{aligned}(\forall x. f(x, y)) \wedge (\forall x. g(x, z)) &= (f(0, y) \wedge f(1, y)) \wedge (g(0, z) \wedge g(1, z)) \\ &= \forall x.(f(x, y) \wedge g(x, z))\,.\end{aligned}$$

### (e) (3%)

The ($\Leftarrow$) direction is correct. To see this, note that

$$(\forall x.f(x,y)) \vee (\forall x.g(x,z)) = f(0,y)f(1,y) \vee g(0,z)g(1,z)\,,$$

while

$$\begin{aligned}\forall x.(f(x,y) \vee g(x,z)) &= (f(0,y) \vee g(0,z)) \wedge (f(1,y) \vee g(1,z)) \\ &= f(0,y)f(1,y) \vee f(0,y)g(1,z) \vee g(0,z)f(1,y) \vee g(0,z)g(1,z) \\ &= (\forall x.f(x,y)) \vee (\forall x.g(x,z)) \vee f(0,y)g(1,z) \vee g(0,z)f(1,y)\,,\end{aligned}$$

which is clearly implied by the first one.

A counterexample for the ($\Rightarrow$) direction is $f = x$ and $g = \neg x$. We have $\forall x.(f(x,y) \vee g(x,z)) = \forall x.(1) = 1$, while $\forall x.f(x,y) \vee \forall x g(x,z) = 0 \vee 0 = 0$.

## 3 [Boolean Function Representation] (9%)

### (a) (3%)

$$\begin{aligned}f = {} & x_1x_2x_3 + x_1x_2x_4 + x_1x_2x_5 + x_1x_3x_4 + x_1x_3x_5 \\ & + x_1x_4x_5 + x_2x_3x_4 + x_2x_3x_5 + x_2x_4x_5 + x_3x_4x_5\,.\end{aligned}$$

### (b) (3%)

$$\begin{aligned}f = {} & (x_1+x_2+x_3)(x_1+x_2+x_4)(x_1+x_2+x_5)(x_1+x_3+x_4)(x_1+x_3+x_5) \\ & (x_1+x_4+x_5)(x_2+x_3+x_4)(x_2+x_3+x_5)(x_2+x_4+x_5)(x_3+x_4+x_5)\,.\end{aligned}$$

### (c) (3%)

The ROBDD is shown in Fig. 1.

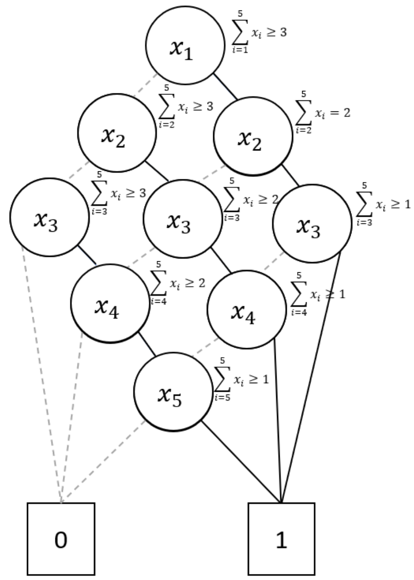


Fig. 1: ROBDD for $f(x_1, \ldots, x_5)$. The variable order is $x_1, x_2, x_3, x_4, x_5$.

**(d) (3%)**

The AIG is shown in Fig. 2.

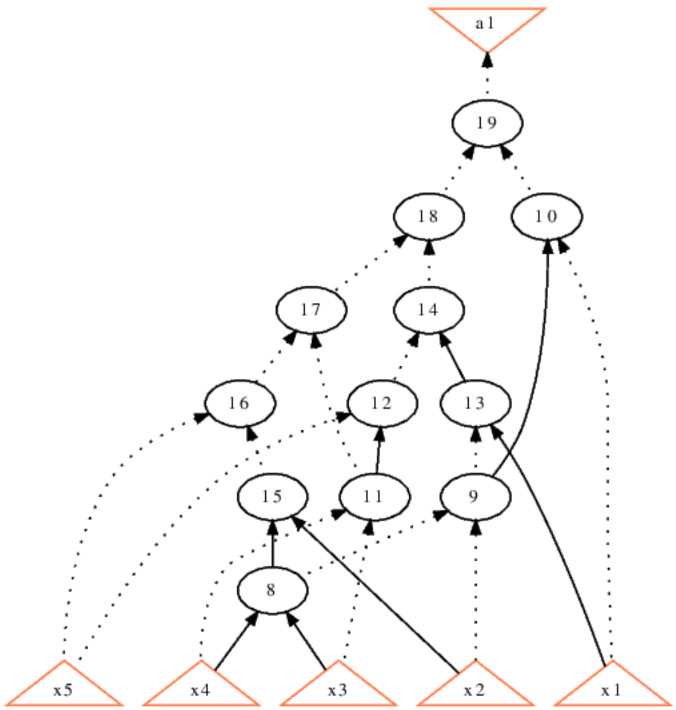


Fig. 2: AIG for $f(x_1, \ldots, x_5)$. The inverted edges are denoted by dotted edges.

### (e) (3%)

$$
\begin{aligned}
f = M(& \\
& M(M(x_1, x_2, x_3), M(x_1, x_2, x_4), M(x_1, x_2, x_5)), \\
& M(M(x_1, x_3, x_4), M(x_1, x_3, x_5), M(x_1, x_4, x_5)), \\
& M(M(x_2, x_3, x_4), M(x_2, x_3, x_5), M(x_2, x_4, x_5)) \\
& ) .
\end{aligned}
$$

## 4 [Application of Quantified Boolean Formula] (16%)

### (a) (8%)

Grading criteria:

- Quantifier: 2%
- Rule constraints: 3%
- Winning condition: 1%
- Correctly assembling everything: 2%

For $(i,j) \in S = \{(1,1),(1,2),(2,1),(2,2),(3,1)\}$, let $x_{t,i,j}$ denote that player $P_X$ take the stone $j$ of the pile $i$ at move $t$ for $t \in \{1,3,5\}$, and let $y_{t,i,j}$ denote that player $P_X$ take the stone $j$ of the pile $i$ at move $t$ for $t \in \{2,4\}$.

We write $X_t$ to denote the set $\{x_{t,i,j} \mid (i,j) \in S\}$ and similarly $Y_t = \{y_{t,i,j} \mid (i,j) \in S\}$ to ease notation.

In addition, we write

$$\textsc{ExactlyOne}(v_1,\ldots,v_k) = \bigvee_{r\in\{1,\ldots k\}} (v_r \wedge \bigwedge_{s\in\{1,\ldots k\}\setminus\{r\}} v'_s)$$

to denote the Boolean function that exactly one of $v_1,\ldots,v_k$ is true.

The QBF is

$$\exists X_1, \forall Y_2, \exists X_3, \forall Y_4, \exists X_5.\, \phi_{X,1} \wedge (\phi_{Y,2} \to (\phi_{X,3} \wedge (\phi_{Y,4} \to (\phi_{X,5} \wedge \phi_W)))) \,,$$

where

$$\phi_{X,1} = \textsc{ExactlyOne}(x_{1,1,1} \vee x_{1,1,2}, x_{1,2,1} \vee x_{1,2,2}, x_{1,3,1}) \,,$$

$$\begin{aligned}\phi_{Y,2} = \textsc{ExactlyOne}&(y_{2,1,1} \vee y_{2,1,2}, y_{2,2,1} \vee y_{2,2,2}, y_{2,3,1}) \\ &\wedge \bigwedge_{(i,j)\in S} x_{1,i,j} \to y'_{2,i,j} \,,\end{aligned}$$

$$\begin{aligned}\phi_{X,3} = \textsc{ExactlyOne}&(x_{3,1,1} \vee x_{3,1,2}, x_{3,2,1} \vee x_{3,2,2}, x_{3,3,1}) \\ &\wedge \bigwedge_{(i,j)\in S} (x_{1,i,j} \vee y_{2,i,j}) \to x'_{3,i,j} \,,\end{aligned}$$

$$\begin{aligned}\phi_{Y,4} = [\textsc{ExactlyOne}&(y_{4,1,1} \vee y_{4,1,2}, y_{4,2,1} \vee y_{4,2,2}, y_{4,3,1}) \\ &\vee (\bigwedge_{(i,j)\in S} y'_{4,i,j} \wedge \bigwedge_{(i,j)\in S} (x_{i,i,j} \vee y_{2,i,j} \vee x_{3,i,j}))] \\ &\wedge \bigwedge_{(i,j)\in S} (x_{1,i,j} \vee y_{2,i,j} \vee x_{3,i,j}) \to y'_{4,i,j} \\ &,\end{aligned}$$

$$\begin{aligned}\phi_{X,5} = [\textsc{ExactlyOne}&(x_{5,1,1} \vee x_{5,1,2}, x_{5,2,1} \vee x_{5,2,2}, x_{5,3,1}) \\ &\vee (\bigwedge_{(i,j)\in S} x'_{5,i,j} \wedge \bigwedge_{(i,j)\in S} (x_{i,i,j} \vee y_{2,i,j} \vee x_{3,i,j} \vee y_{4,i,j}))] \\ &\wedge \bigwedge_{(i,j)\in S} (x_{1,i,j} \vee y_{2,i,j} \vee x_{3,i,j} \vee y_{4,i,j}) \to x'_{5,i,j} \,,\end{aligned}$$

and

$$\phi_W = (\bigvee_{(i,j)\in S} x_{3,i,j} \wedge \bigwedge_{(i,j)\in S} y'_{4,i,j}) \vee \bigvee_{(i,j)\in S} x_{5,i,j} \,.$$

$\phi_{X,i}$ and $\phi_{Y,i}$ encode the conditions that $P_X$ and $P_Y$ must satisfy according to the rules, and the first player to violate the rules loses immediately. If all moves are valid, $\phi_W$, which encodes the winning condition of $P_X$, decides which player wins.

### (b) (8%)

Grading criteria:

- True/$P_X$ wins: 3%
- Strategy: 5% (Since the problem statement is ambiguous, giving strategy in terms of the original game or in terms of QBF are both okay.)

The formula is true. $P_X$ is the winning player. A winning strategy is

$$\begin{cases} f_{1,1,1} = 0 \\ f_{1,1,2} = 0 \\ f_{1,2,1} = 0 \\ f_{1,2,2} = 0 \\ f_{1,3,1} = 1 \\ f_{3,1,1} = y_{2,2,1} \vee y_{2,2,2} \\ f_{3,1,2} = y_{2,2,1} \wedge y_{2,2,2} \\ f_{3,2,1} = y_{2,1,1} \vee y_{2,1,2} \\ f_{3,2,2} = y_{2,1,1} \wedge y_{2,1,2} \\ f_{3,3,1} = 0 \\ f_{5,1,1} = y_{2,1,2} \wedge y_{4,2,2} \\ f_{5,1,2} = ((y_{2,1,1} \vee y_{2,2,1}) \wedge y_{4,2,2}) \vee (y_{2,2,2} \wedge y_{4,2,1}) \\ f_{5,2,1} = y_{2,2,2} \wedge y_{4,1,2} \\ f_{5,2,2} = ((y_{2,1,1} \vee y_{2,2,1}) \wedge y_{4,1,2}) \vee (y_{2,1,2} \wedge y_{4,1,1}) \\ f_{5,3,1} = 0\,. \end{cases}$$

## 5 [Binary Decision Diagram] (16%)

### (a) (8%)

See Theorem 1 of [1].

Grading criteria:

- Base case: 3%
- Considering reduction rules: 2%
- Proving equivalence given that no reduction rules can be applied: 3%

**(b) (8%: 4% each (2% ordering, 2% explanation))**

The variable order $x_1, y_1, x_2, y_2, \ldots, x_n, y_n$ results in an ROBDD with $2n+2$ nodes, since each variable only has one corresponding decision node. Part of the BDD is shown in Fig. 3.

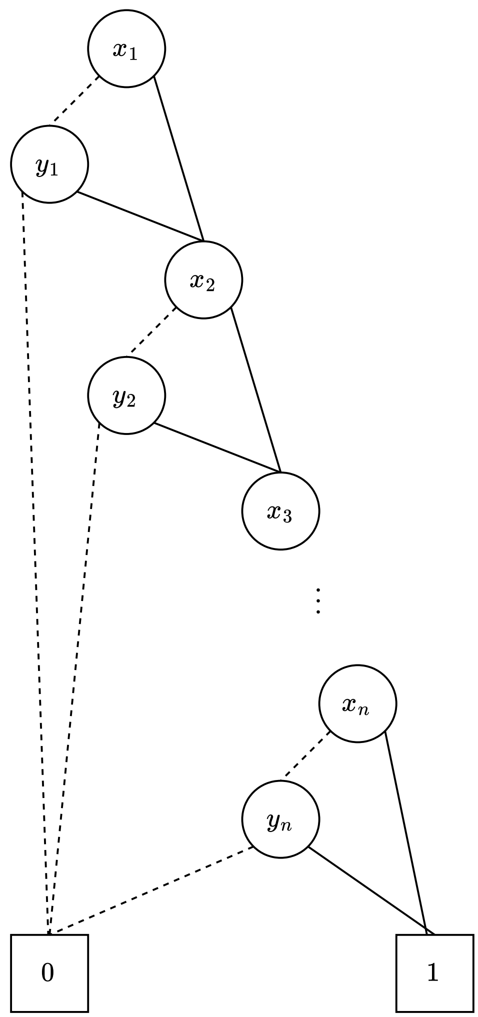


Fig. 3: A BDD with $O(n)$ nodes for problem 5(b).

On the other hand, the variable order $x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_n$ results in an ROBDD with at least $2^n$ nodes.[1] To see this, we consider any two minterms

[1] We interpret the problem as asking for an ordering with $\Omega(2^n)$ nodes. Otherwise, the previous ordering suffices, as $O(n) \subset O(2^n)$.

$m_1 \neq m_2$ over $x_1, \ldots, x_n$. WLOG assume that $x_i \in m_1$ and $x_i' \in m_2$. Then, the assignment $y_1, \ldots, y_{i-1}, y_i, y_{i+1}, \ldots, y_n = 1, \ldots, 1, 0, 1, \ldots, 1$ will satisfy $f_{m_1}$ but not $f_{m_2}$, and thus $f_{m_1} \neq f_{m_2}$. Therefore, the top half of the BDD corresponds to a complete binary tree of height $n-1$, which has $2^n - 1$ nodes. Part of the BDD is shown in Fig. 4.

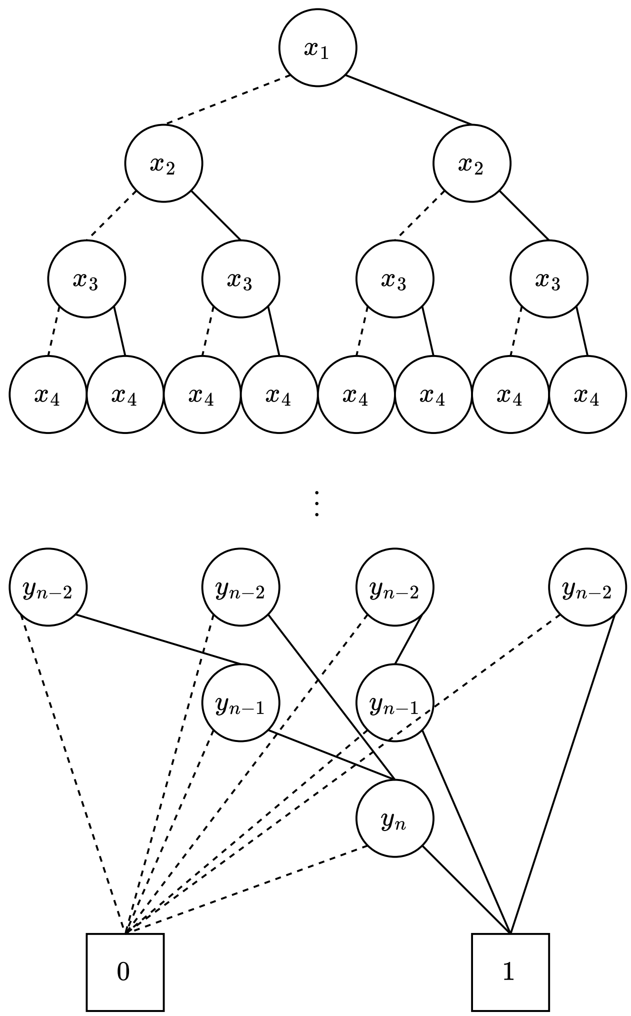


Fig. 4: A BDD with $O(2^n)$ nodes for problem 5(b).

## References

[1] Randal E. Bryant. "Graph-Based Algorithms for Boolean Function Manipulation". In: *IEEE Trans. Comput.* 35.8 (Aug. 1986), pp. 677–691. ISSN: 0018-9340. DOI: 10.1109/TC.1986.1676819.